

**ENHANCEMENT OF PGENRO INFORMATION MANAGEMENT SYSTEM
WITH AUTOMATED DOCUMENT CLASSIFICATION
AND DIGITAL REPOSITORY**

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May 2026

Chapter I

INTRODUCTION

In the modern digital era, organizations rely on Information Management Systems (IMS) to efficiently manage large volumes of data and records. These systems help organize, store, retrieve, and process information, improving office efficiency while reducing manual work and possible errors. In government offices, an effective IMS is important because documents must be properly recorded, monitored, secured, and accessed to support daily administrative operations.

The Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province uses an Information Management System (IMS) to manage and record different office documents and transactions. As part of its daily operations, the office handles various incoming and outgoing documents, letters, and other official papers, including travel orders, office memoranda, employee records, inventories, service requests, Inventory Custodian Slips (ICS), and visitors' logbooks. These records are important in supporting office coordination, environmental management activities, documentation processes, and the monitoring of official transactions within the office.

Based on the initial interview and observation conducted by the researchers, PGENRO already has an existing Information Management System. However, the current IMS is a desktop-based application that uses Microsoft Access as its database. Since it is installed only on specific computers, not all staff in the office can directly access or use the system. As a result, documents from different sections are still submitted to the Administrative Division of PGENRO for manual encoding, recording, storage, and processing.

The existing Information Management System also has several limitations. It has no built-in charts and analytics for monitoring office transactions, and it does not have its own internal storage or secure digital repository. At present, documents received from Gmail or physical copies are uploaded to Google Drive using a single link outside the IMS, and the link is shared through Microsoft Teams. This setup makes document tracking, retrieval, organization, and security more difficult because file storage is not fully integrated into the system.

Furthermore, the current process still depends on manual encoding, which may cause human errors, duplicate entries, and slower retrieval of information. In addition, the visitors' logbook is still handwritten and manually encoded into the system, which increases the workload of personnel and may lead to delays, inconsistent entries, and difficulty in monitoring visitor records efficiently. Requests for system upgrades also take several months or even years before action is implemented by the Provincial Information and Communications Technology Office of Quezon Province. These limitations affect office productivity and show that the existing Information Management System needs improvement.

Therefore, enhancing the existing Information Management System of PGENRO is necessary. This study aims to transform the current desktop-based Information Management System into a modern web-based IMS with automated document classification and a secure digital repository. The proposed system will be developed specifically for PGENRO to centralize the management of incoming and outgoing documents, letters, and other official papers while improving the storage, monitoring, retrieval, and security of office records. The enhanced system will support better document

organization, faster access to records, improved document tracking, charts and analytics, reduced manual encoding, and more efficient information management within the office.

Project Context

Effective information management is essential for organizations that handle large volumes of records, documents, and administrative transactions. Government offices, in particular, require organized and reliable systems to properly manage official records, monitor office operations, and maintain accurate documentation of transactions and activities. A well-managed Information Management System (IMS) helps improve office efficiency, supports faster retrieval of records, strengthens document security, and reduces delays caused by manual processing and record handling.

The Provincial Government Environment and Natural Resources Office (PGENRO) currently uses an Information Management System (IMS) to manage and record official documents, records, and administrative transactions within the office. As part of its daily operations, the office processes and records different official documents and records, including incoming and outgoing documents, letters, and other administrative papers such as travel orders, office memoranda, employee records, inventory records, service requests, Inventory Custodian Slips (ICS), and visitors' logbook entries. These records are important in supporting office coordination, monitoring environmental activities, maintaining documentation, and supporting the daily administrative functions of the office.

The Provincial Government Environment and Natural Resources Office (PGENRO) currently uses a desktop-based Information Management System (IMS) that

utilizes Microsoft Access as its database. While the existing system provides basic support for recording and managing office documents, it remains limited in accessibility, storage integration, document tracking, and monitoring features. Since the system is installed only on specific office computers, not all authorized personnel can directly access or use the system. As a result, several documents and records from different office sections are still submitted to the Administrative Division for manual encoding, recording, storage, and processing.

Based on the researchers' interview and observation, the current Information Management System has several operational limitations. The system does not have built-in charts and analytics for monitoring office transactions and document activities. In addition, it does not have its own centralized digital repository for securely storing electronic copies of documents within the IMS itself. At present, documents received from Gmail or physical copies are uploaded separately to Google Drive, while file links are shared through Microsoft Teams outside the IMS environment. Because document storage is not fully integrated into the system, document tracking, retrieval, organization, and security become more difficult to manage efficiently.

Furthermore, some office processes still rely on manual encoding and partially manual document management processes. The visitors' logbook, for instance, is still handwritten and manually encoded into the system, which increases the workload of office personnel and may result in delays, inconsistent entries, duplicate records, and human errors. Requests for system upgrades and improvements often take several months or even years before implementation. These limitations affect office productivity and demonstrate the need to improve the existing Information Management System of PGENRO.

These conditions establish the need to enhance the existing Information Management System of PGENRO. Although the office already uses an IMS, the current system does not fully address the growing demands of document management, storage, retrieval, monitoring, accessibility, and security. The issue is not the absence of a system, but the limitations of the existing desktop-based IMS and the continued reliance on manual and external processes. As the volume of office records and transactions continues to increase, there is a need for a more modern, centralized, accessible, and secure Information Management System.

In relation to the needs of the office, the proposed system will include features that directly address the limitations of the existing IMS. The system will allow authorized users to record, upload, classify, search, and retrieve incoming and outgoing documents, letters, and other official papers more efficiently. It will also include automated control number generation, document tracking, audit trails, charts and analytics, and a centralized digital repository to improve monitoring, accessibility, and security of office records. In addition, the proposed system will provide role-based user authentication to ensure that only authorized personnel can access sensitive information and official documents. These features are based on the problems identified through the researchers' interview and observation, such as limited accessibility, manual encoding, external file storage, lack of analytics, handwritten visitor logbooks, and difficulty in tracking and retrieving records.

To address these challenges, this study proposes the Enhancement of the PGENRO Information Management System with Automated Document Classification and Digital Repository. The study aims to transform the current desktop-based IMS into a modern web-based Information Management System capable of improving document storage,

monitoring, retrieval, accessibility, and security. The proposed system will include automated document classification, centralized digital repository storage, automated control number generation, advanced search and filtering, charts and analytics, and secure role-based user authentication.

Through this enhancement, the proposed system is expected to reduce manual workload, minimize human errors, improve document organization, strengthen record security, and provide a more efficient and reliable information management process for the Provincial Government Environment and Natural Resources Office of Quezon Province.

Table 1. Thematic Table for key Issues

Theme	Key Issue	Interview Snippet
Limited System Accessibility	The current Information Management System is only used by six personnel from the Administrative Division	<i>“In the current system we are using, only the six of us in the Administrative Division are able to access and use the system.”</i> (PGENRO CDA II, 2026)
Lack of Integrated Digital Repository	The current system does not have an integrated digital repository. Instead, Google Drive links are used for storing uploaded documents, and the links are shared through Microsoft Teams.	<i>“The current system uses Google Drive links for file storage, and these links are shared through our Microsoft Teams group where we upload the documents encoded in the system.”</i> (PGENRO CDA II, 2026)

Delayed System Enhancement	System enhancements and requested improvements take several months or even years before implementation.	<i>“The enhancement of the current system may take several months or even years once it is requested from PICTO. (PGENRO CDA II, 2026)</i>
Outdated Database System	The current Information Management System uses an outdated Microsoft Access database, which may limit system accessibility, scalability, security, and overall performance.	<i>“The system we are currently using is only connected to a Microsoft Access database, which is already outdated for the database being used in this system.” (PGENRO CDA II, 2026)</i>
Manual Encoding Errors	Manual encoding may result in duplicate entries and typographical errors.	<i>“In manual encoding, there are times when duplicate entries occur, and sometimes typographical errors happen during the encoding of records.” (PGENRO CDA II, 2026)</i>

Purpose and Description

This capstone project aims to enhance the existing Information Management System (IMS) of the Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province by transforming the current desktop-based system into a modern web-based Information Management System with Automated Document Classification and Digital Repository. The proposed enhancement aims to improve the efficiency of document and record management within the office by reducing manual

encoding processes, minimizing human errors, improving document accessibility, and providing a more organized and secure platform for storing and retrieving office records.

The proposed system is specifically designed to support the operational needs of PGENRO in handling and managing incoming and outgoing documents, letters, and other official papers such as travel orders, office memoranda, employee records, inventory records, service requests, Inventory Custodian Slips (ICS), and visitors' logbook entries. Through the enhanced system, authorized personnel will be able to manage, monitor, search, retrieve, and store records more efficiently and securely.

The enhanced Information Management System will include the following capabilities:

1. Automatically generates unique control numbers for encoding documents to prevent duplication and improve document tracking.
2. Allow authorized users to upload, store, organize, and retrieve scanned and electronic copies of official documents through a secure centralized digital repository. Supports automated document classification to help categorize and organize uploaded documents more efficiently within the system.
3. Provides advanced search and filtering features for faster retrieval and monitoring of records and office transactions.
4. Includes charts and analytics for monitoring office document activities and transactions, particularly incoming and outgoing documents.
5. Implements role-based user authentication and access control to ensure that only authorized personnel can access sensitive office records and system functions.

6. Maintains digital recording and monitoring of visitors' logbook entries to reduce manual processes and improve record management efficiency.
7. Provides audit trails and document tracking features to help monitor system activities and document transactions.

The proposed web-based Information Management System is expected to improve document organization, strengthen record security, reduce manual workload, minimize encoding errors, and support faster and more efficient information management processes within Provincial Government Environmental Natural and Resources Office of Quezon Province.

The Enhancement of the Information Management System with Automated Document Classification and Digital Repository is expected to benefit the following:

- **PGENRO Staff.** The enhanced PGENRO Information Management System addresses the challenges of manual document handling by automating classification, organizing records, and providing a secure digital repository. This improvement is especially valuable in offices managing large volumes of communications, inventories, and service requests. By minimizing encoder errors, reducing the risk of document loss, and enabling faster retrieval of information, staff can devote more time to their core responsibilities, such as monitoring environmental programs and supporting administrative operations. Studies support the adoption of digital information systems to enhance operational efficiency, data integrity, and security, ultimately benefiting both the office and its personnel.

- **Future Researchers.** The enhanced PGENRO Information Management System provides a foundation for future studies by demonstrating how automated document classification and digital repositories can improve office efficiency and data management. Researchers can build upon this system to explore advanced features such as full artificial intelligence-driven document analysis, predictive reporting, or integration with other government agencies. By studying and enhancing digital information systems, future researchers can contribute to the development of more efficient, secure, and scalable solutions for public administration, ultimately supporting better decision-making and service delivery in government offices.
- **Students.** The enhanced PGENRO Information Management System benefits students by providing a practical example of how digital technologies can be applied to real-world organizational problems. Through this capstone, students gain hands-on experience in web-based system development, automated document classification, and secure digital repositories. By studying the system's design, functionality, and implementation, students can enhance their technical skills, analytical thinking, and problem-solving abilities. Research highlights that engaging with applied information systems projects helps students bridge theoretical knowledge with practical experience, preparing them for future careers in technology, information management, and related fields.

The system will also support functions such as recording and managing incoming and outgoing documents, maintaining inventories, monitoring service requests, and

managing administrative records within the office. By integrating these functions into a single system, the enhanced Information Management System will streamline information processing, reduce manual workload, and improve the accuracy and accessibility of records.

Objectives of the Study

This study generally aims to enhance the existing Information Management System (IMS) of the Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province by transforming the current desktop-based system into a modern web-based Information Management System with Automated Document Classification and Digital Repository. Specifically, this study aims to:

1. Develop a secure web-based Information Management System that:
 - 1.1 manage, store, search, retrieve, and monitor PGENRO office records through a centralized digital repository.
 - 1.2 read and extract text from uploaded PDF files and scanned copies using Optical Character Recognition (OCR).
 - 1.3 classify documents through automated and user-assisted categorization;
 - 1.4 generate unique control numbers, provide charts and data analytics, and implement role-based access control for secure and efficient document management.

- 1.5 provides digital recording and monitoring of visitors' logbook entries, including graphical monitoring and reporting features for visitor records and office transactions.
2. Test the functionality of the system using Blackbox testing.
3. Evaluate the level of acceptability of the system based on ISO 25010 standards:
 - 3.1. Functional Suitability
 - 3.2. Performance Efficiency
 - 3.3. Interaction Capability
 - 3.4. Combability
 - 3.5. Reliability
 - 3.6. Security
 - 3.7. Maintainability
 - 3.8. Flexibility
 - 3.9. Safety

Scope and Limitation

The Enhanced PGENRO Information Management System (IMS) focuses on transforming the current desktop-based Information Management System of the Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province into a modern web-based system with Automated Document Classification and Digital Repository. The system is designed to improve the management, storage, retrieval, monitoring, and security of office records, documents, and administrative transactions within PGENRO.

The system covers the management of incoming and outgoing documents, letters, and other official papers, including travel orders, office memoranda, employee records, inventory records, service requests, Inventory Custodian Slips (ICS), and visitors' logbook entries. It allows authorized users to record, upload, store, search, retrieve, view, and monitor scanned and electronic copies of documents through a centralized digital repository.

The system includes Optical Character Recognition (OCR) technology to read and extract text from uploaded PDF files and scanned copies of documents. This supports automated and user-assisted document classification, allowing uploaded documents to be categorized according to document type and related information. It also includes automated control number generation to prevent duplication, advanced search and filtering, charts and analytics, document tracking, audit trails, and role-based user authentication to ensure secure and efficient document management.

The system is limited to the operations of PGENRO Quezon Province only. It does not support direct integration with other government offices, including CENROs, DENR, or other external agencies. The system also does not include predictive analytics, self-learning artificial intelligence models, or automatic decision-making features. While OCR and automated classification are included, the accuracy of classification may still depend on the quality of uploaded PDF files or scanned copies and the correctness of user-provided information.

Additionally, the system does not include and does not support full offline functionality. It requires a stable internet connection for full operation, and system performance may depend on the reliability of the office's internet infrastructure. Future

enhancements such as mobile application development, integration with other government systems, advanced AI prediction, and full offline synchronization are beyond the scope of this study.

This is aligned with your objective may centralized digital repository, OCR, automated/user-assisted classification, control numbers, charts/analytics, and role-based access.

Definition of Terms

The following terms are operationally defined to provide clarity and understanding of their usage within this study:

Audit Trail refers to the system record of user activities such as logging in, uploading, editing, deleting, or retrieving documents.

Automated Document Classification refers to the system feature that categorizes uploaded documents based on document type or extracted text.

Control Number refers to a unique number automatically generated by the system for tracking incoming and outgoing documents.

Digital Repository refers to the centralized storage where scanned and electronic copies of official documents are securely uploaded, organized, stored, and retrieved.

Enhanced Information Management System refers to the proposed modern web-based system designed to improve the existing desktop-based IMS of PGENRO.

Incoming and Outgoing refer to letters, official papers, and records received or released by PGENRO, including travel orders, office memoranda, service requests,

inventory records, ICS, and other administrative documents.

Information Management System (IMS) refers to the existing system used by PGENRO to manage, record, store, and retrieve official documents, records, and administrative transactions.

Optical Character Recognition (OCR) refers to the technology used to read and extract text from uploaded PDF files and scanned copies of documents.

Role-Based Access Control refers to the security feature that limits system access based on the user's assigned role, such as administrator or staff.

Web-Based System refers to a system that can be accessed through a web browser by authorized users using an internet connection.

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Chapter II

REVIEW OF LITERATURE

This chapter presents a comprehensive summary and analysis of existing research related to the enhancement of the PGENRO information management system with automated document classification and digital repository. It serves as the foundation for the study by providing an overview of established concepts, technologies, and systems relevant to information management system. The review highlights key findings, identifies gaps in existing systems, and explores various approaches in improving document handling processes. Through the analysis of related literature, researchers gain insights into current trends, challenges, and future directions in the development of intelligent and efficient information management system.

Theoretical Framework

This study is anchored on Information Systems Theory, Records Continuum Theory, and the Technology Acceptance Model (TAM). These theories support the development of the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository by explaining how information systems improve organizational operations, how records should be managed throughout their life cycle, and how users accept and utilize technology in the workplace.

The Information Systems Theory explains that an information system is composed of people, processes, data, hardware, software, and networks that work together to collect, process, store, and distribute information for organizational decision-making. In the

context of the **Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province**, this theory supports the enhancement of the existing system by converting manual and locally stored processes into a more organized and accessible web-based platform. Through the proposed system, incoming and outgoing communications, travel orders, office memoranda, employee records, inventory, service requests, visitor logs, and Inventory Custodian Slips can be properly managed in one centralized system. This theory emphasizes that an effective information system improves accuracy, efficiency, accessibility, and security in handling organizational records.

The study is also supported by the Records Continuum Theory, which views records management as a continuous process from the creation, capture, organization, storage, use, and preservation of records. This theory is applicable to the proposed system because the office handles various types of documents that must be properly classified, stored, retrieved, and protected. Through the digital repository, documents can be stored in a centralized database, reducing the risk of misplaced or lost files. The automated document classification feature also supports this theory by helping authorized personnel categorize documents more consistently and efficiently. Instead of relying only on manual sorting, the system assists in organizing digital records based on document type and purpose.

Furthermore, the Technology Acceptance Model (TAM) serves as a foundation for understanding the acceptance of the proposed system by its intended users. TAM explains that users are more likely to adopt a system if they perceive it as useful and easy to use. In this study, the proposed information management system is designed to provide a user-friendly interface, secure login, advanced search, document upload, automated classification, and role-based access. These features aim to make document management

easier for staff and administrators. If users find the system helpful in reducing manual work, improving document retrieval, and preventing data loss, they are more likely to accept and use the system effectively.

These theories collectively guide the development of the proposed system. Information Systems Theory provides the basis for improving office operations through technology. Records Continuum Theory supports the proper management and preservation of documents in a digital environment. The Technology Acceptance Model ensures that the system will be designed according to user needs, usability, and usefulness. Together, these theories strengthen the foundation of the study by showing how the enhanced system can improve document handling, classification, storage, retrieval, and overall information management.

Information Management System

An Information Management System (IMS) is a system used to collect, organize, store, process, and retrieve information to support an organization's daily operations and decision-making. It helps institutions manage records more systematically, improve the flow of information, and reduce delays caused by manual handling of documents and data. Studies in recent years continue to show that information systems are important in improving work efficiency, data accessibility, and administrative coordination.

In the study "Information management system for research of Don Mariano Marcos Memorial State University–South La Union Campus" by Malaya, Munar, Cuison, and Dacanay (2022), the researchers developed an information management system intended to improve the management of research records in a university setting. The study used a

descriptive-developmental approach and applied the Rapid Application Development model, with evaluation based on ISO/IEC 25010. Its respondents included faculty members and research office personnel, and the study showed that the developed system helped provide more organized handling of research information and supported institutional record management more effectively.

Automated Documents Classification

Automated document classification is a process that uses technologies such as Optical Character Recognition (OCR), machine learning, and Natural Language Processing (NLP) to identify, extract, and categorize documents based on their content. It helps organizations manage large volumes of digital and scanned files more efficiently by reducing manual sorting, improving document organization, and making records easier to search and retrieve.

In the study “Automatic PDF Document Classification with Machine Learning” by Llàcer Luna, Garigliotti, Martínez-Plumed, and Ferri Ramírez (2024), the researchers developed an automated classification pipeline for the Universitat Politècnica de València’s Alfresco document repository, which contained around 600,000 PDF files, but only about 100,000 were correctly categorized. The study used OCR and machine learning to distinguish scanned and digital documents, extract text, and classify the files into 51 predefined categories using models such as BERT and Random Forest. The study showed that automated document classification can improve document organization, accessibility, and retrieval, which is highly related to the proposed Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository because it supports the need for automatic sorting, faster searching, and systematic storage

of office documents.

Digital Repository

A digital repository is a system used to store, organize, preserve, and retrieve digital documents in a centralized electronic location. It helps organizations manage important files such as memoranda, reports, employee records, travel orders, and scanned documents in a secure and accessible environment. Through a digital repository, institutions can reduce paper-based filing, prevent document loss, and improve the speed of retrieving records for daily operations.

In the study “AI-based Integrated Approach for the Development of Intelligent Document Management System (IDMS)” by Pandey, Arora, Arora, Goyal, Gera, and Yadav (2023), the researchers developed an intelligent document management system that integrates artificial intelligence, machine learning, Optical Character Recognition (OCR), and Natural Language Processing (NLP) to improve document handling and storage. The study emphasized that a digital repository plays an important role in ensuring that classified and processed documents are securely stored and easily retrievable when needed. The findings showed that centralized digital storage improves workflow efficiency, document accessibility, and overall record management. This study supports the importance of using a digital repository in managing large volumes of documents and maintaining secure, organized, and accessible records in an institution.

Innovation solution of Information Management System

The current use of PGENRO Information Management System (IMS) is application-based, which limits its accessibility, scalability, and overall efficiency. To

make the system better and more responsive to modern organizational needs, the proposed innovation is to transform it into a more advanced and modern Information Management System with web-based functionality and Artificial Intelligence (AI) features.

Current Information Management System (IMS) of PGENRO Quezon Province



Figure 1. Existing Desktop-Based PGENRO Information Management System Interface

The collaboration between the researchers and the Provincial Government Environment and Natural Resources Office (PGENRO) focuses on enhancing the existing Information Management System into a more efficient, accessible, and web-based platform with Artificial Intelligence (AI) features. To achieve this enhancement, the researchers follow a systematic development process.

The process starts with data gathering through interviews, observations, and consultations with PGENRO staff to identify the limitations of the existing desktop-based Information Management System. During this phase, the researchers identify issues such as limited accessibility, manual encoding, external Google Drive storage, absence of charts and analytics, and dependence on Microsoft Access database.

After gathering the requirements, system analysis is conducted to evaluate the

current workflow, document handling process, and office transactions. The researchers analyze how documents are received, encoded, stored, and retrieved within the office. This phase helps determine the needed enhancements for the proposed system.

Next, the system design phase is conducted to create the structure of the enhanced IMS, including the database design, user interface, dashboards, document classification process, digital repository structure, and role-based access. During this phase, the researchers also design the integration of Firebase database, secure authentication, analytics dashboard, and automated document classification.

After the design phase, system development is implemented using Visual Studio Code, web technologies, and Firebase services. The researchers develop features such as document uploading, automated document classification, secure login and authentication, digital repository management, advanced search and filtering, charts and analytics, and document tracking.

Once development is completed, testing and validation are conducted using Black Box Testing and user acceptance testing with PGENRO personnel to evaluate the functionality, reliability, security, and performance of the enhanced system. Necessary revisions and improvements are applied based on the feedback gathered from users.

Finally, deployment and evaluation are conducted to assess the effectiveness of the enhanced IMS in improving document management, reducing manual workload, increasing accessibility, and supporting more efficient office operations within PGENRO.

System Requirement

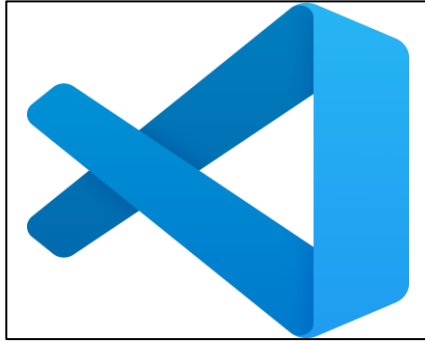


Figure 2. Visual Studio Code

Source: <https://quarto.org/docs/get-started/images/vscode-logo.png>

Visual Studio Code is used as the main development environment in enhancing the PGENRO Information Management System. According to Del Sole (2023), Visual Studio Code supports different development scenarios, including web, mobile, cloud, Python programming, and AI-related services. This makes it suitable for the proposed system because the study focuses on enhancing the existing desktop-based IMS into a more accessible web-based platform with automated document classification and a digital repository.

In this study, VS Code supports the development of the proposed system using web technologies such as HTML, CSS, JavaScript, and Python. These technologies are needed to create the user interface, manage system functions, connect the system to the database, and support automation features. Since the existing IMS uses Microsoft Access and has limited accessibility, VS Code provides a more flexible environment for developing a centralized and scalable system that can improve the current process of PGENRO.

VS Code also helps the developers build and test important features of the enhanced Information Management System, such as secure login and authentication, document

uploading, automated document classification, digital repository management, advanced search and filtering, and dashboard monitoring with charts and analytics. These features are connected to the main goal of the study, which is to improve the accessibility, organization, storage, and retrieval of documents within PGENRO.

Furthermore, VS Code is useful in the development process because it supports debugging, extensions, and version control. These tools help the developers identify errors, improve the system functions, and manage project files properly during development. Through VS Code, the researchers are able to create a system that responds to the limitations of the existing IMS, particularly its desktop-based setup, manual encoding process, lack of internal storage, absence of charts, and limited accessibility for office personnel.

Therefore, Visual Studio Code is appropriate for this study because it supports the technical requirements of enhancing the PGENRO IMS. It helps in developing a web-based, organized, secure, and efficient information management system that provides automated document classification, centralized digital repository, better document tracking, and improved support for the daily operations of PGENRO.

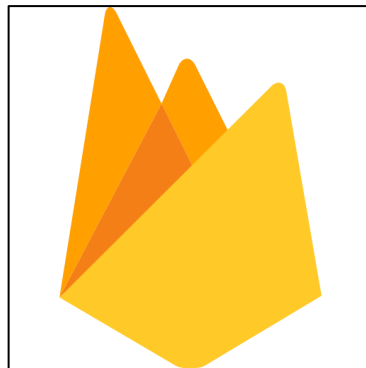


Figure 3. Firebase

Source: https://cdn.iconicons.com/icons2/2699/PNG/512/firebase_logo_icon_171157.png

Firebase is a cloud-based development platform provided by Google that supports web and mobile application development through services such as database management, authentication, analytics, and cloud storage. According to the official Firebase documentation, Firebase provides Cloud Firestore and Realtime Database, which support real-time data synchronization and centralized data storage across connected users and devices. It also provides Firebase Authentication for secure user sign-in and identity management.

Madaminov and Allaberganova (2023) explain that Firebase is useful in modern application development because it supports cloud-based structure, real-time synchronization, and efficient data handling. Their study highlights that Firebase supports applications that require updated records, secure storage, and fast data retrieval.

In relation to this study, Firebase supports the enhancement of the PGENRO Information Management System by providing a centralized and cloud-based database for storing and managing document-related records. Since the existing IMS uses Microsoft Access and operates as a desktop-based application, Firebase supports the development of a more accessible web-based system. Through Firebase, authorized personnel access synchronized records, uploaded documents, and transaction data using internet-connected devices.

Firebase also supports important features of the proposed enhanced IMS, including user authentication, role-based access, document storage, and real-time record updating. These features are relevant because the existing IMS has limitations in accessibility, centralized storage, and document management. In addition, Firebase supports the digital repository feature by organizing uploaded documents and maintaining secure and centralized storage of records.

Furthermore, Firebase stores document classifications, transaction logs, and dashboard data used for charts and analytics within the system. This is relevant to the proposed enhancement because the existing IMS does not provide built-in analytics and relies on external storage platforms such as Google Drive links shared through Microsoft Teams. Therefore, Firebase is relevant to this study because it supports centralized information management, secure storage, synchronized access, and improved document handling for the enhanced Information Management System of the PGENRO.



Figure: 4 Hypertext Markup Language (HTML)

Source: <https://cdn-icons-png.flaticon.com/512/1532/1532556.png>

HTML is used in this study as one of the main technologies for developing the user interface of the enhanced PGENRO Information Management System. Rashmi, Baba, and Bhagwan (2021) explain that Hypertext Markup Language is a basic programming language used in creating web pages and organizing web content. This supports the need of the study because the proposed system is enhanced from a desktop-based IMS into a web-based platform.

In relation to this study, HTML helps structure the pages and content of the enhanced IMS, including the login page, dashboard, document upload page, digital repository page, record management page, and user management page. It provides the

foundation for arranging text, forms, buttons, tables, links, and other interface elements needed by authorized PGENRO personnel.

HTML is also relevant to the proposed system because it supports accessibility and usability in a web-based environment. Since the existing IMS is desktop-based and can only be accessed on selected computers, HTML helps create web pages that users can access through browsers using internet-connected devices. This supports the goal of improving system accessibility for authorized PGENRO staff.

Furthermore, HTML works together with CSS, JavaScript, Firebase, and Python-related features to support the functions of the enhanced IMS. While HTML provides the structure of the web pages, other technologies handle the design, interactivity, database connection, authentication, document storage, and automated document classification. Therefore, HTML is relevant to this study because it supports the development of a web-based interface for managing documents, accessing records, uploading files, and using the digital repository of the enhanced PGENRO Information Management System.



Figure. 5 Cascading Style Sheets (CSS)

Source: <https://logowik.com/content/uploads/images/css-icon5555.logowik.com.webp>

The CSS (Cascading Style Sheets) is a web technology used to control the visual

presentation and layout of web pages. It is responsible for the design of elements such as colors, fonts, spacing, alignment, and responsive structure, making websites more organized and user-friendly. Jain et al. (2024) explained that CSS, together with HTML, plays an important role in modern web development because it improves web design and functionality through proper styling practices. In addition, Bhanarkar et al. (2023) emphasized that responsive web design relies on CSS definitions such as media queries and flexible layouts so that web pages can adjust to different screen sizes and provide a better user experience. In the study titled “Enhancement of PGENRO Information Management System with Automated Document Classification and Digital Repository,” CSS is relevant because it can help create a clear, organized, and responsive user interface for document encoding, repository browsing, file viewing, and information retrieval, thereby improving the usability and accessibility of the proposed web-based system.

We will use CSS (Cascading Style Sheets) in the enhancement of our Information Management System because it is essential for designing a clean, professional, and user-friendly interface. While HTML provides the structure of the web-based platform, CSS allows us to control the visual presentation—such as layout, colors, fonts, and spacing—which makes the system easier to navigate and more appealing to users. Since one of our goals is to improve efficiency and usability for PGENRO staff, CSS ensures that forms, dashboards, and charts are not only functional but also visually organized, reducing confusion and helping users quickly locate the information they need. Another important reason is responsiveness: CSS enables the system to adapt to different screen sizes, so authorized personnel can access records seamlessly whether they are using a desktop, laptop, or mobile device. By applying CSS, we can maintain consistency in design across

all pages of the system, highlight important data through visual cues, and create a modern interface that aligns with the standards of a secure web-based application. In short, CSS is necessary because it transforms the system from being just functional into being intuitive, accessible, and visually professional, which directly supports our capstone's goal of enhancing usability and efficiency.

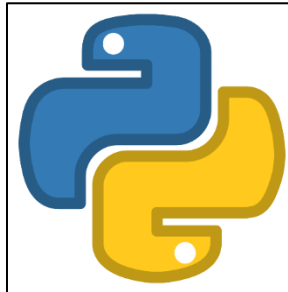


Figure. 6 Python

Source: <https://sl.bing.net/hrZLDjjmqi>

The Python programming language is a high-level programming language widely used in system development because of its readability, flexibility, and broad support for web development, automation, data processing, and artificial intelligence. Rayhan and Gross (2023) explained that Python has become increasingly popular in research and development because it is easy to learn, versatile, and applicable in several domains such as web development, data science, machine learning, natural language processing, robotics, and scientific computing. This shows that Python is a practical programming language for developing modern information systems that require both standard system functions and intelligent features.

We will use Python in the enhancement of our Information Management System because it is one of the most versatile and powerful programming languages, especially for implementing advanced features like automated document classification and AI-driven

data processing. Unlike HTML and CSS, which handle structure and design, Python allows us to integrate intelligent functionalities such as Optical Character Recognition (OCR), Natural Language Processing (NLP), and machine learning models that can read, analyze, and categorize scanned documents automatically. This directly supports our goal of reducing manual encoding and minimizing human errors in the system.

Python also works seamlessly with Firebase as our chosen database, enabling smooth data storage, retrieval, and synchronization in real time. Its wide range of libraries and frameworks makes it easier to build secure authentication systems, generate analytics, and implement audit trails for transparency. Another advantage is Python's strong community support and extensive documentation, which ensures that our development team can troubleshoot issues and continuously improve the system. By using Python, we can ensure that the enhanced IMS is not only web-based but also intelligent, scalable, and future-ready, capable of handling large volumes of documents while maintaining efficiency and accuracy.

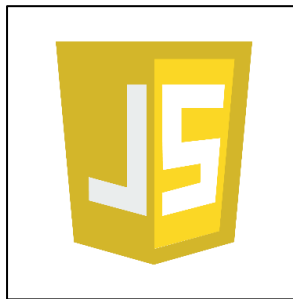


Figure. 7 JavaScript

Source: <https://sl.bing.net/h2UvuaIGgYS>

JavaScript is a high-level, versatile programming language widely used in developing interactive and dynamic web applications. It is known for its flexibility, event-driven features, and compatibility with modern browsers, making it useful in the development of client-side and server-side systems. According to Hamaamin (2023),

JavaScript remains one of the most widely used programming languages because of its adaptability, ease of use, and ability to create efficient and maintainable applications. The study also emphasized that JavaScript supports software development in different fields due to its flexibility and strong development environment.

JavaScript is relevant because it can be used in developing system components that require responsive user interfaces, efficient processing, and organized data handling. Its flexible nature allows developers to design modular and maintainable programs, which is important in creating an information management system with multiple functions such as document processing, classification support, and repository management. Therefore, JavaScript can contribute to the development of a dependable and efficient system that supports the improvement of document management practices in PGENRO.

We will use JavaScript in the enhancement of our Information Management System because it is the key technology that brings interactivity and dynamic functionality to a web-based platform. While HTML provides the structure and CSS handles the design, JavaScript allows us to make the system responsive and interactive for PGENRO staff. With JavaScript, we can implement features such as real-time data validation, dynamic dashboards, automated chart generation, and responsive document tracking, which are essential for monitoring incoming and outgoing communications. It also enables seamless integration with Firebase, allowing real-time updates to the database and ensuring that records are instantly synchronized across all authorized users. Another important reason is that JavaScript supports frameworks and libraries like React, Angular, or Vue, which can help us build a more efficient and user-friendly interface. By using JavaScript, we ensure that the enhanced IMS is not just static but interactive, responsive, and capable of handling

complex workflows, making document management faster and more reliable. In short, JavaScript is necessary because it transforms the system into a dynamic web application that supports automation, analytics, and real-time interaction, directly aligning with our capstone's goal of creating a modern, secure, and efficient information management solution.

Black box testing

Black box testing is a software testing method that examines a system's functionality based on inputs and outputs without considering its internal code. According to the ISTQB Foundation Level Syllabus, black-box techniques such as equivalence partitioning and boundary value analysis are used to derive test cases from requirements and expected behavior. Studies support its usefulness in evaluating software quality. Alshamaa (2025) reported through a literature review that black box testing remains effective in assessing test efficiency and effectiveness across systems, while Santi et al. (2022) found that black box testing successfully identified functional defects in an academic information system. Therefore, black box testing is suitable for this study because it helps determine whether the Enhanced PGENRO Information Management System performs its intended functions accurately and reliably.

ISO 25010 Evaluation

The International Standard ISO 25010 provided a framework for evaluating the quality of software systems. The criteria used included Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, Portability, and

Interaction Capability. These parameters were essential in assessing the effectiveness of the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository. By ensuring that the system meet these standards, the Provincial Government Environment and Natural Resources Office (PGENRO) could maintain a reliable, secure, and user-friendly web based platform for document classification, storage, and retrieval.

Functional Suitability

Functional suitability refers to the degree to which a system provides functions that meet stated and implied needs under specified conditions. According to ISO/IEC 25010, it is an important quality characteristic used to assess whether a software product delivers correct, complete, and appropriate functions based on user requirements. Nugraha and Rachmana (2024) stated that functional suitability is essential in evaluating whether a web-based application effectively supports user tasks and meets user expectations. In this study, functional suitability ensures that the Enhanced PGENRO Information Management System provides accurate and appropriate functions for document encoding, classification, storage, and retrieval.

Performance Efficiency

Performance efficiency refers to the degree to which a system provides appropriate performance relative to the amount of resources used under specified conditions. This quality characteristic focuses on how fast and efficiently the system responds to user requests, processes data, and utilizes available resources such as memory, processing time, and storage. In software evaluation, performance efficiency is important because it

determines whether the system can perform its intended functions smoothly and without unnecessary delays.

In the study, performance efficiency ensures that the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository can handle document encoding, classification, searching, storage, and retrieval in a timely and efficient manner. The system is expected to provide fast response time, stable performance, and efficient resource utilization during operation. This quality factor is essential in determining whether the developed system can support the daily processes of the Provincial Government Environment and Natural Resources Office (PGENRO) effectively and efficiently.

Interaction Capability

Interaction capability refers to the degree to which the system can effectively communicate and interact with users, other systems, and external platforms. For the Enhanced PGENRO Information Management System, this parameter ensures that staff can interact with the system through intuitive interfaces, responsive dashboards, and real-time feedback mechanisms. It emphasizes smooth communication between the user and the application, allowing tasks such as document encoding, classification, and retrieval to be performed seamlessly.

Beyond user interaction, this capability also supports interoperability with external systems, such as cloud storage services, email platforms, or future government agency databases. By enabling integration and data exchange, the system can expand its functionality and remain adaptable to evolving organizational needs. Interaction capability also includes accessibility features, ensuring that users with varying levels of technical

expertise can operate the system effectively. In this study, strengthening interaction capability promotes transparency, accountability, and collaboration, while ensuring that the Enhanced IMS remains responsive, user-centered, and future-ready.

Compatibility

Compatibility refers to the ability of the system to operate in different environments and integrate with other applications or platforms. For the Enhanced IMS, compatibility ensures that the web-based system can run across multiple browsers and devices, while also supporting integration with existing office workflows. It also emphasizes interoperability, allowing the system to handle diverse file formats, interact with external applications such as email platforms or cloud storage, and remain functional as technology evolves. This characteristic reduces deployment issues, supports scalability, and ensures smooth adoption by PGENRO staff.

Reliability

Reliability refers to the system's ability to perform consistently under specified conditions. For the Enhanced IMS, reliability ensures that document classification, storage, and retrieval functions operate without failure, even during high workloads. Backup and recovery mechanisms, audit trails, and version control strengthen reliability by protecting data and ensuring that the system remains dependable in supporting PGENRO's daily operations.

Security

Security measures the system's ability to protect information and data from unauthorized access, breaches, or corruption. For the Enhanced IMS, security is achieved

through role-based authentication, controlled user permissions, and secure storage in the digital repository. Audit trails and encryption mechanisms safeguard sensitive office records, ensuring confidentiality, integrity, and accountability. This parameter is essential in maintaining trust and compliance with government standards.

Maintainability

Maintainability refers to the ease with which the system can be modified, updated, or improved. For the Enhanced IMS, maintainability ensures that developers can efficiently update features, fix errors, and enhance functionalities without disrupting operations. Modular design, clear documentation, and scalable architecture support long-term sustainability, allowing the system to adapt to future requirements and technological advancements.

Flexibility

Flexibility refers to the system's ability to adapt to changes in requirements, environments, or workflows without requiring extensive redevelopment. For the Enhanced PGENRO Information Management System, flexibility ensures that the platform can accommodate future upgrades such as integration with other government agencies, expansion of automated classification features, or the addition of mobile support. This characteristic is important because organizational needs evolve over time, and a flexible system reduces the cost and effort of adjustments. By designing the IMS with modular architecture and scalable components, the system can respond to new demands while maintaining stability and efficiency. Flexibility guarantees that the system remains relevant and sustainable in the long term.

Safety

Safety refers to the system's ability to operate without causing harm to users, data, or organizational processes. In the context of the Enhanced IMS, safety ensures that sensitive records are protected from accidental loss, corruption, or unauthorized exposure. Features such as backup and recovery mechanisms, audit trails, and controlled access contribute to safe operations by minimizing risks associated with human error or technical failures. Safety also includes ensuring that the system does not compromise the integrity of environmental and administrative records, which are critical for government decision-making. By prioritizing safety, the Enhanced IMS provides a secure environment where staff can confidently manage documents without fear of data breaches or operational disruptions.

The reviewed literature shows that Information Management Systems (IMS) automated document classification and digital repositories are becoming more important. They help organizations work efficiently especially those with a lot of records. Many studies agree that modern information systems are better than ways of managing data.

Studies on IMS say they help organizations work smoothly and make better decisions. For example IMS reduces the workload, errors and helps find information faster (Malaya et al. 2022). This supports the need to improve the PGENRO system, which still uses manual encoding and limited-access desktop applications.

Automated document classification is also important. Technologies like Optical Character Recognition (OCR) machine learning and Natural Language Processing (NLP) help process. Categorize many documents. A study by Ll  cer Luna et al. (2024) Found that automated classification improves document organization, accessibility and retrieval. This

is relevant to PGENRO, which currently sorts documents manually and has classification.

Digital repositories provide a secure and accessible way to store electronic documents. Pandey et al. (2023) Said that integrating repositories with intelligent document management systems improves workflow efficiency, data integrity and document accessibility. This supports the idea of integrating a repository in the enhanced PGENRO system.

The literature also highlights the importance of development tools and technologies. Cloud-based platforms and scalable databases are useful. Firebase enables real-time data synchronization, secure authentication and efficient cloud storage (Madaminov and Allaberganova 2023). Python is good for implementing features like automated classification and data processing (Rayhan & Gross 2023).

Web development technologies like HTML, CSS and JavaScript are important for creating responsive and user-friendly systems. These technologies support interactive and efficient web-based platforms (Rashmi et al. 2021; Hamaamin, 2023; Jain et al. 2024).

When evaluating a system Black Box Testing is often used to assess functionality based on input-output behavior (Alshamaa, 2025; Santi et al., 2022). The ISO/IEC 25010 standard provides a framework for evaluating software quality.

Overall the literature supports the development of a web-based Information Management System (IMS) with automated document classification and a digital

repository. The studies justify the need to enhance PGENRO by adopting technologies and centralized systems leading to improved efficiency, reduced manual processes, better data management and enhanced organizational performance. These insights provide a foundation, for designing and implementing the proposed enhanced IMS.

Conceptual Framework

The conceptual framework illustrates the input-process-output (IPO) model, which outlines the procedures and processes involved in the enhancement and operation of the information management system.

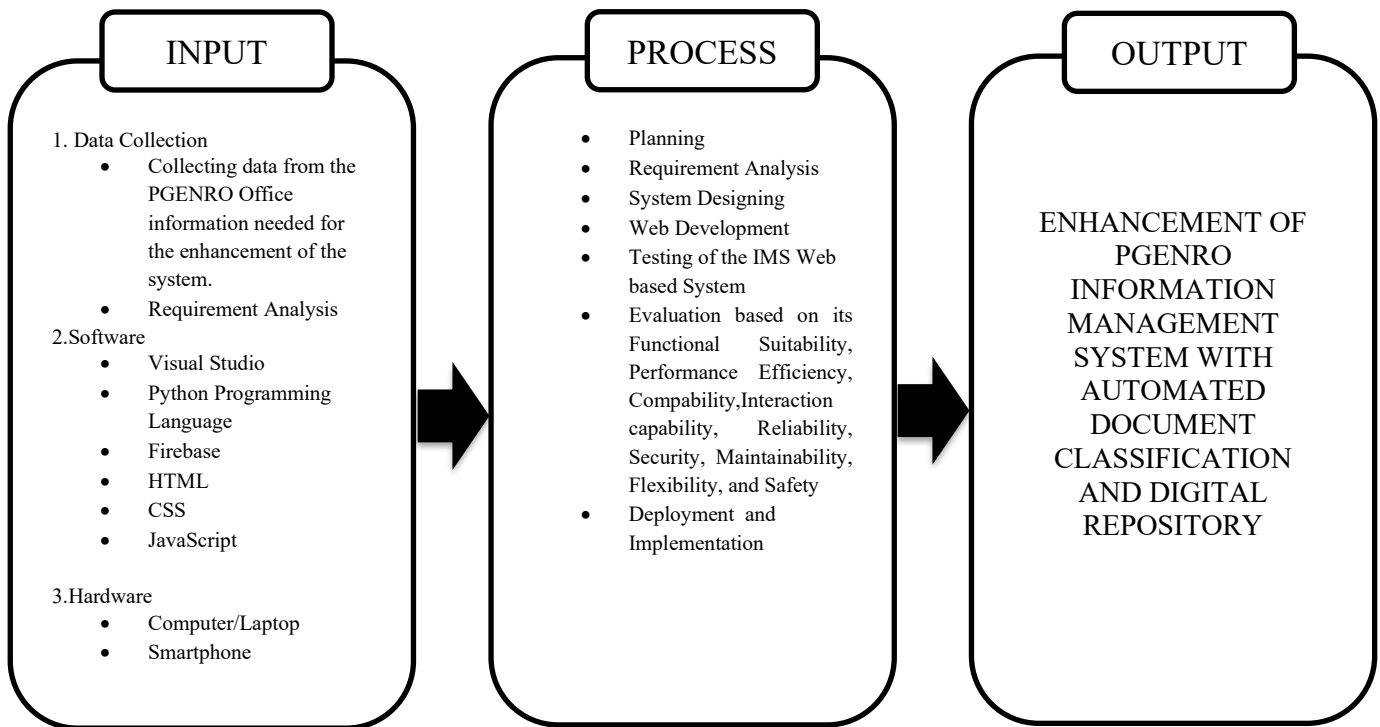


Figure: 8 Conceptual Framework of the study

In Figure 8, the researchers present the conceptual framework that guided the innovation and development of the proposed system. The study aims to enhance the

existing Information Management System of the Provincial Government Environment and Natural Resources Office (PGENRO) by transforming the current desktop-based process into a secure Web-Based Information Management System with Automated Document Classification and Digital Repository.

The input phase includes the existing PGENRO Information Management System, software and hardware requirements, and the identified problems encountered in the current system. These problems include limited system accessibility, manual encoding, external file storage through Google Drive links shared in Microsoft Teams, outdated Microsoft Access database usage, lack of document tracking, absence of charts and reporting features, handwritten visitors' logbook entries, and delayed system enhancement requests. The input phase also includes the tools and technologies needed for the development of the proposed system.

The process phase involves requirement analysis, system planning, system design, web-based system development, integration of Optical Character Recognition (OCR), automated and user-assisted document classification, implementation of a centralized digital repository, testing, and evaluation. This phase explains how the researchers analyzed the existing office workflow and transformed the identified problems into system features and enhancements that support efficient document management.

The output phase is the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository. The proposed system is designed to provide a reliable, secure, accessible, and efficient platform for managing, storing, searching, retrieving, tracking, and monitoring office records and transactions. It also includes features such as control number generation, charts and reporting tools, role-

based access control, audit trails, and digital monitoring of visitors' logbook entries. Through this conceptual framework, the study clearly demonstrates the relationship between the identified office problems, the development process, and the expected system output.

Chapter III

METHODOLOGY

This chapter will describe the research design, methods, and procedures that will be adopted in the study. It will explain how the researchers will gather the necessary data, identify the respondents, utilize the research instruments, and carry out the system evaluation. It will also include the ethical considerations that will be observed throughout the conduct of the capstone project. Through this methodological framework, the study will be guided in a clear and organized manner, ensuring that the development and assessment of the proposed system will be aligned with the objectives of the research.

Research Locale

The study will be conducted at the Provincial Government Environment and Natural Resources Office, located in Lucena City. The office is responsible for managing environmental programs, natural resource administration, document processing, inventory monitoring, employee records, and other administrative functions within the province of Quezon.

The researchers will select this office as the research locale because it is the primary institution using the existing Information Management System (IMS) that requires enhancement. The current system is application-based and limited to office computers, which creates challenges in document accessibility, storage, retrieval, and monitoring. Issues such as duplicate records, scattered file storage, manual encoding, and difficulty in document tracking are expected to serve as the basis for identifying the needed improvements in office operations.

Since the study will focus on the enhancement of the existing Information Management System through automated document classification and digital repository

integration, the Provincial Government Environment and Natural Resources Office will serve as the most appropriate setting for the implementation and evaluation of the proposed capstone project

Respondents

This study will involve three groups of respondents providing a broader evaluation of the proposed Enhancement of PGENRO Information Management System with Automated Document Classification and Digital Repository.

For the performance testing, the respondents will be composed of five (5) IT expert from Southern Luzon State University Lucena Campus. They will be selected to evaluate the technical performance of the system in terms of functionality, accuracy, efficiency, reliability, and overall system operation. Through Black Box Testing, they will assess whether the system produces the expected results based on the given inputs and required functions.

For the acceptability testing, the respondents will consist of six (6) administrative staff and twenty (20) staff personnel from the Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province. They will be chosen because they are the intended users of the system and are directly involved in handling, encoding, storing, retrieving, and monitoring office documents and records.

For the ISO/IEC 25010 evaluation, the same group of respondents, composed of five (5) IT experts, six (6) administrative staff, and twenty (20) PGENRO staff personnel, will evaluate the system based on Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Flexibility,

and Safety.

By involving IT experts, administrative staff, and PGENRO staff personnel, the study will be able to examine the system's technical performance, level of acceptability, and software quality in the actual working environment.

Research Instrument

The primary research instrument that will be used in this study will be the Pass-Fail survey instrument for Black Box Testing. This instrument will be utilized to assess the functionality of the proposed system by focusing on its input and output behavior without examining its internal code structure. Through this instrument, the researchers will be able to determine whether the system will perform according to the expected requirements under different test conditions. Various test cases will be prepared based on possible user actions and system scenarios, and the outputs produced by the system will be examined to identify if they are correct, reliable, and consistent. Since Black Box Testing emphasizes the external performance of the system, it will serve as an appropriate tool in verifying whether the developed application meets its intended functions and specifications.

Additionally, the researchers will use an adapted 4-point Likert scale questionnaire to measure the acceptability of the proposed system based on the ISO/IEC 25010 Software Quality Model. The instrument will be administered to the respondents to evaluate the system in terms of Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Maintainability, Portability, and Flexibility. These criteria will be used to assess the overall quality and user acceptance of the developed system. The scale will be interpreted as follows: (4) Highly Acceptable,

(3) Acceptable, (2) Unacceptable, and (1) Highly Unacceptable. Through this questionnaire, the respondents will be able to rate each criterion based on their experience and evaluation of the system. The use of this instrument will help the researchers determine the level of acceptability of the application and provide supporting data for the overall assessment of the developed capstone project.

Data Gathering Procedure

A formal letter of request, together with the research adviser's endorsement, is prepared by the researchers and submitted to the Provincial Government Environment and Natural Resources Office (PGENRO) of Quezon Province for approval to conduct the study. Upon receiving approval, the researchers proceed with the gathering of necessary data related to the existing Information Management System used by the office.

To strengthen the foundation of the study, the researchers gather information from related literature, journals, and published and unpublished studies relevant to information management systems, automated document classification, digital repositories, and ISO/IEC 25010 software quality evaluation. These references support the analysis, design, development, and evaluation of the proposed capstone project.

The researchers also conduct interviews and observations with authorized personnel and staff to identify the current processes, challenges, and requirements in managing organizational records and documents. The collected information serves as the basis for the design and development of the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository.

After the development of the proposed system, Black Box Testing is conducted to evaluate the functionality and performance of the application. The respondents then assess

the acceptability of the system using the adapted ISO/IEC 25010-based questionnaire. The gathered data are analyzed and interpreted using the Arithmetic Weighted Mean (AWM) to determine the level of acceptability of the developed system.

Work Plan

Table 1 show the work plan of the study, which will present the sequence of activities to be undertaken by the researchers from planning, analysis, design, and development up to testing and finalization of the proposed system. It will provide a systematic framework that will guide the conduct of the study and the completion of the project titled “Enhancement of PGENRO Information Management System with Automated Document Classification and Digital Repository.” Through this work plan, the researchers will be able to organize the necessary tasks, monitor the progress of the study, and ensure the timely completion of each research phase. Furthermore, it will illustrate the overall flow and development of the capstone project.

Table 1

Schedule of Activities

Activities	Mar. 2026	Apr. 2026	May. 2026	jun. 2026	Jul. 2026	Aug. 2026	Sept. 2026	Oct. 2026	Nov. 2026
1. Planning and conceptualizing research for the study proposal									
2. Data gathering for documentation									
3. Submission of proposal									
4. Checking, revision, and finalization of research documents									
5. Designing, developing, programming, and debugging of the Enhanced PGENRO Information Management System									
6. Testing of the application									
7. Implementation and evaluation of the application using ISO/IEC 25010 and Black Box Testing									
8. Final evaluation and preparation of manuscript									

System Development

Table 1 illustrate the sequential process of the study, covering the major phases from planning, data gathering, system design, development, testing, implementation, and

final evaluation. The planning stage will involve collecting necessary information and identifying the problems encountered in the existing PGENRO Information Management System, which will serve as the foundation of the proposed enhancement. Through interviews, observations, and document analysis, the researchers will gather relevant data needed to support the development of the study.

After the data collection phase, the researchers will proceed to the design stage, where the system flow, database structure, interface layout, and functional modules of the proposed system will be prepared. This will include the design of automated document classification, digital repository, control number generation, and reporting features. During the development phase, the researchers will focus on coding, programming, debugging, and integrating all system components to ensure smooth operation and functionality.

Once the development is completed, the system will undergo testing and validation to evaluate its performance, reliability, and usability using Black Box Testing and ISO/IEC 25010 software quality standards. Necessary revisions and corrections will be made based on the results of the evaluation to improve the overall performance of the system. Finally, the implementation and final evaluation phase will ensure that the enhanced system will be ready for deployment, followed by the preparation of the final manuscript and presentation for the capstone defense.

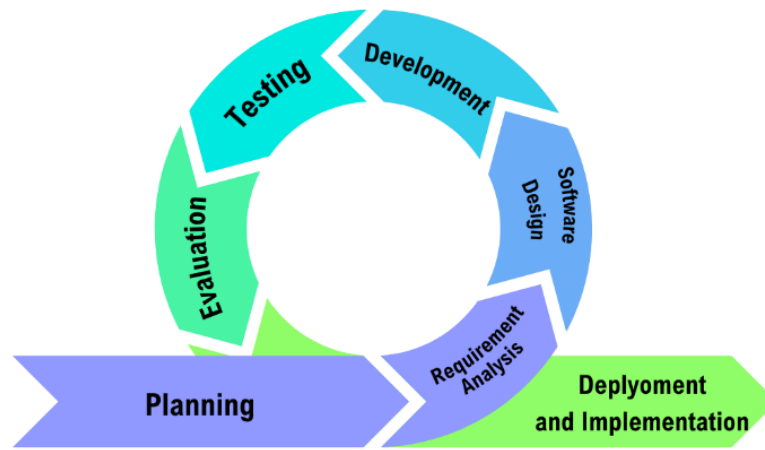


Figure 9. Agile Diagram

Finally, the researchers proceed to the evaluation stage to assess the effectiveness of the Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository. Presented below are the different stages involved in the development of the proposed system.

1. Planning

The system development for this research project involves a detailed plan to create a secure, efficient, and user-friendly Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository. The researchers utilize key software tools to ensure the system's effectiveness in managing office documents and records. Development is carried out using Visual Studio Code for coding and system development, while the user interface is implemented using HTML, CSS, and JavaScript to provide an interactive and accessible platform for users.

Firebase is used for database management, user authentication, cloud storage, and real-time data synchronization to ensure secure and organized record

handling. Python is used for the artificial intelligence component of the system, particularly in reading, analyzing, and classifying uploaded documents based on their content. The development process includes planning, software selection, system design, development, testing, deployment, and maintenance, resulting in a reliable and adaptable system that improves document storage, retrieval, and overall workflow efficiency in PGENRO.

2. Requirement Analysis

The Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository will aim to improve document handling and records management at the Provincial Government Environment and Natural Resources Office (PGENRO) by addressing inefficiencies in manual filing, document retrieval, and record monitoring. This section will analyze the current technical situation and the proposed system, detailing the hardware, software, functional, and user requirements needed for the successful development and implementation of the system. The current information management system in PGENRO relies mostly on manual handling of documents by office personnel. The challenges encountered include time-consuming document filing and retrieval, possible misplacement or duplication of records, limited centralized storage for digital copies, manual classification of documents, and difficulty in monitoring and accessing important files efficiently.

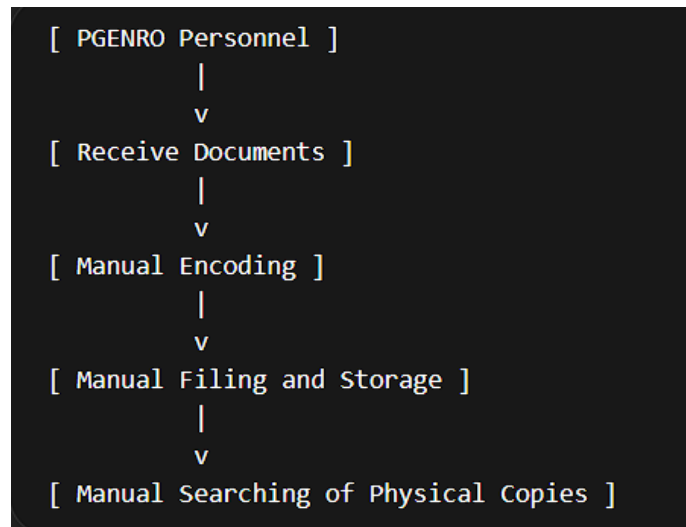


Figure 10. Current workflow

The current Information Management System in PGENRO relies mostly on manual handling of documents by office personnel. The challenges expected to be addressed will include time-consuming document filing and retrieval, possible misplacement or duplication of records, limited centralized storage for digital copies, manual classification of documents, and difficulty in monitoring and accessing important files efficiently.

Figure 10 show the current information management workflow in PGENRO. The identified issues will include manual document encoding, physical filing and storage, slow retrieval of records, possible misplacement or duplication of files, and the absence of automated document classification and centralized digital storage.

- Proposed Enhance Information Management System

The proposed Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository will improve document management by using a web-based platform with Firebase and Python-based AI support. The system will help reduce manual document handling by allowing authorized users to store, classify, search, and retrieve documents more efficiently.

The features of the proposed system will include automated document classification to reduce manual sorting errors, a digital repository for centralized document storage, a search and retrieval function for faster access to records, user authentication for secure access, report generation for monitoring documents, and real-time data synchronization through Firebase to ensure updated and organized records.

The proposed enhanced IMS workflow will be presented in Figure 11. The expected improvements will include faster document processing through AI-based document reading and classification, reduced errors in manual encoding and filing, centralized storage through a digital repository, easier searching and retrieval of records, improved data security through user access verification, and more organized monitoring of PGENRO documents and transactions.

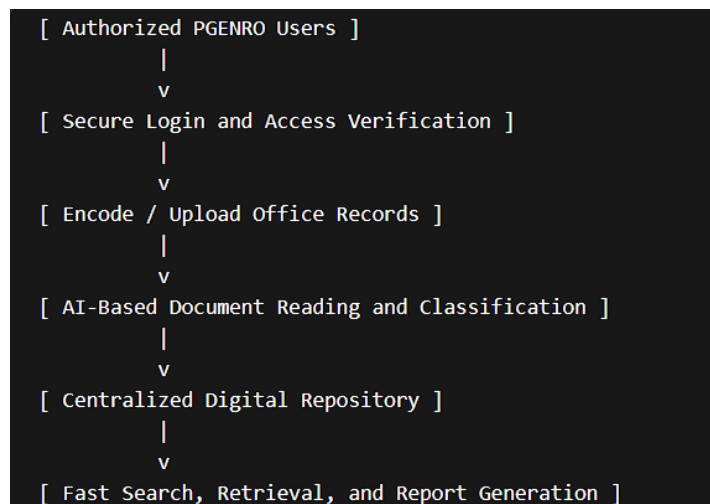


Figure 11. Proposed enhanced IMS

The proposed enhanced IMS workflow is presented in Figure 11. The expected improvements include faster document processing through AI-based document reading and classification, reduced errors in manual encoding and filing, centralized storage through a

digital repository, easier searching and retrieval of records, improved data security through user access verification, and more organized monitoring of PGENRO documents and transactions.

- Technical Requirements

- Software Requirement:

For the development of the proposed Enhanced PGENRO Information Management System, Visual Studio Code will be used for coding and system development. HTML, CSS, and JavaScript will be used for the user interface, while Firebase will serve as the database, authentication, cloud storage, and real-time synchronization platform. Python will be used for the artificial intelligence component, particularly for reading, analyzing, and classifying uploaded documents. The system will require Windows 10 or higher and updated web browsers for compatibility and accessibility.

- Hardware Requirements:

For document storage, system processing, and daily office operations, a computer or laptop will be necessary with a minimum of 8GB RAM, Intel Core i5 8th generation or higher processor, and a stable internet connection for smooth system performance. A scanner or multifunction printer will also be needed for converting physical documents into digital files for uploading and classification. External storage devices may also be used for backup and additional file preservation.

- User Requirements:

For administrative access, authorized PGENRO personnel such as the admin and records staff will manage, upload, classify, and monitor office documents. Staff users will be responsible for encoding records, retrieving files, and generating reports. The users must have basic computer knowledge and proper authorization to ensure secure access and effective use of the system.

The proposed Enhanced PGENRO Information Management System will be intended to improve the way office records are managed, stored, classified, and retrieved. Through the use of AI-based document reading, Firebase storage, user authentication, and a centralized digital repository, the system will help minimize manual work, lessen document misplacement, improve data accessibility, and provide a more organized process for handling PGENRO documents.

The Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository will be a web-based system designed to improve the management of documents and records in the Provincial Government Environment and Natural Resources Office (PGENRO) Quezon Province. It will ensure organized document storage, automatic classification of uploaded files, secure access for authorized personnel, and faster retrieval of records. The system will also provide centralized digital repository storage, report generation, and real-time monitoring of office transactions to support more efficient and reliable office operations.

3. Requirement Documentation

- **System Overview**

The Enhanced PGENRO Information Management System with Automated Document Classification and Digital Repository is a web-based system designed to improve the management of documents and records in the Provincial Government Environment and Natural Resources Office (PGENRO) Quezon Province. It ensures organized document storage, automatic classification of uploaded files, secure access for authorized personnel, and faster retrieval of records. The system also provides centralized digital repository storage, report generation, and real-time monitoring of office transactions to support more efficient and reliable office operations.

4. Functional Requirements

The proposed system will include the following major features

A. User Management

Admin Login and Access Control will provide secure authentication for authorized PGENRO personnel to ensure safe access to the system. Request Access Account will allow PGENRO staff to submit account registration requests for system access, subject to admin approval. Role-Based Permissions will assign different access levels for admin and staff to maintain security and control over sensitive records. Staff Account Management will enable the admin to approve, update, or deactivate staff accounts for proper user management.

B. Records and Transaction

Record Encoding will allow users to encode incoming and outgoing communications, travel orders, office memoranda, employee records, inventory, service requests, and ICS. Visitors Log Management will allow visitors to be recorded digitally and enable staff to monitor visitor information efficiently. Record Updating will allow authorized users to edit and update existing records. Document Monitoring will track and monitor stored records and official transactions.

C. Document Management

Document Uploading will allow users to upload office documents and records. Automated Document Classification will classify uploaded documents based on their content using Python-based AI. Digital Repository Storage will store documents in a centralized and organized digital repository.

D. Reports and Security

Report Generation will generate reports for documents, records, office transactions, and visitor logs. Activity Log will record user actions such as login, upload, update, and file retrieval. Backup and Data Protection will help protect important files from loss, damage, or unauthorized access.

E. Search and Retrieval

The Search Function will enable users to search documents by keyword, category, date, or document type. The Filter Function will allow sorting of records based on document classification or transaction type. View and Download Documents will allow authorized users to access stored files when needed.

- **Storyboard Representation**

Presented below will be the visual representation of how the proposed system will appear once it is fully developed and implemented.

Storyboard: Information Management System interface design

A. Login Page

- User enters login credentials (Admin or Staff)
- Unauthorized users will receive an invalid login notification and may request account access if not yet registered.
- Authorized users are redirected to their designated dashboard based on their access level

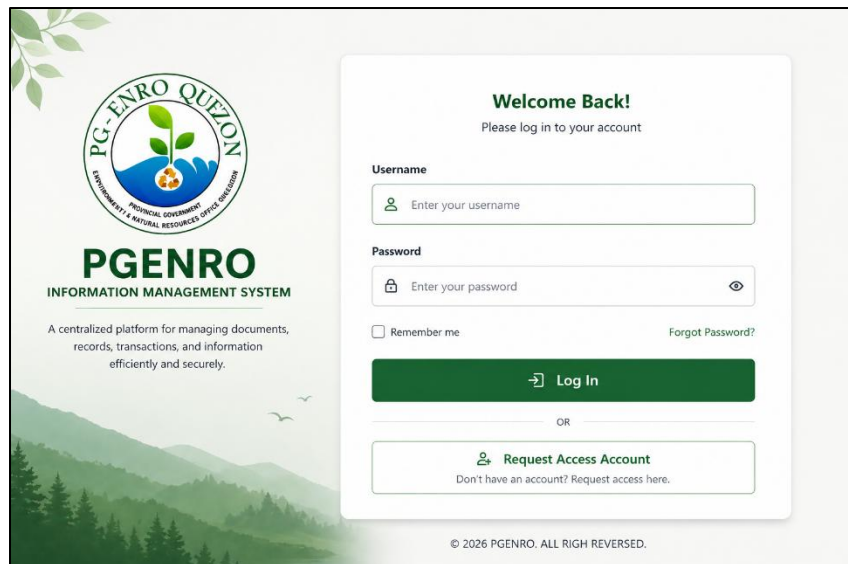


Figure 12. Login Page

A. Admin Dashboard

- Manage incoming and outgoing communications, travel

orders, office memoranda, employee records, inventory, visitor's log, service requests, and ICS records.

- Review and approve request access account submissions from PGENRO staff.
- Edit, update, or delete documents and records under admin access.
- View uploaded documents, office transactions, and pending staff requests.

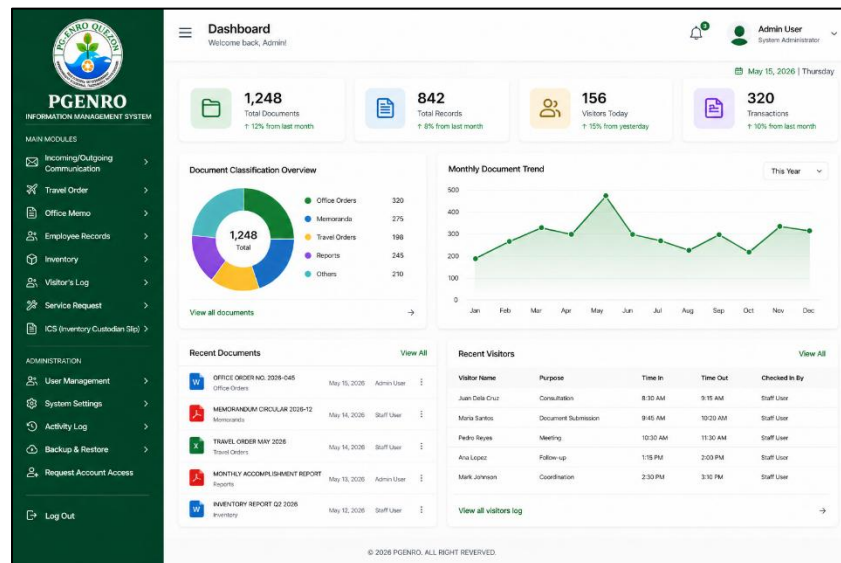


Figure 13. Admin Dashboard

B. User Dashboard

- Upload documents such as travel orders, office memoranda, service requests, inventory records, and other official files.
- Monitor the status of uploaded files, pending requests, and approved records.
- View and monitor visitor log records for tracking visitor

information and office visits.

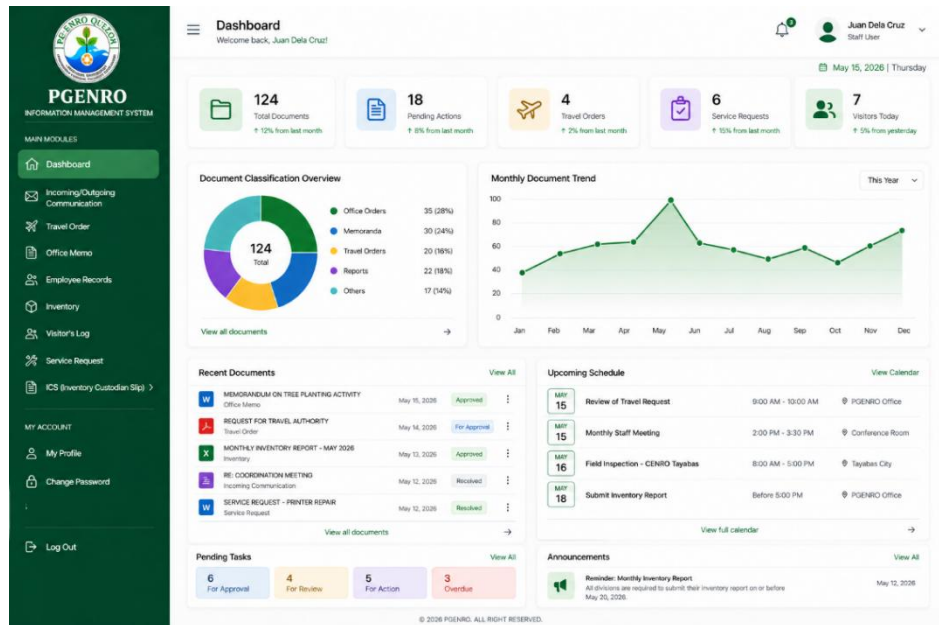


Figure 14. User Interface

- System Functional Summary
 - Generates unique control numbers for documents to prevent duplication.
 - Allows users to upload, view, edit, and monitor documents based on their assigned access level.
 - Improves institutional efficiency by reducing misplaced files, duplicate records, and manual processing errors.
 - Offers interactive dashboards with real-time data, document status, reports, and system activity.
 - Supports major PGENRO modules such as Incoming/Outgoing Communication, Travel Order, Office Memo, Employee Records, Inventory, Visitor's Log, Service Request, and ICS.

4. Software Design

The software design phase will focus on the preparation of the system's structure and workflow for the proposed Enhancement Information Management System of PGENRO. This will include the use of use-case diagrams to illustrate the interaction between the users and the system. It will also cover the design of the system interfaces, dashboards, and major modules that will serve as the basis for the development and implementation of the project.

4.1. Use- Case Diagram

Figure 14 present the use-case diagram of the proposed PGENRO Information Management System, showing the different interactions of the users with the system and the functions they will be able to perform.

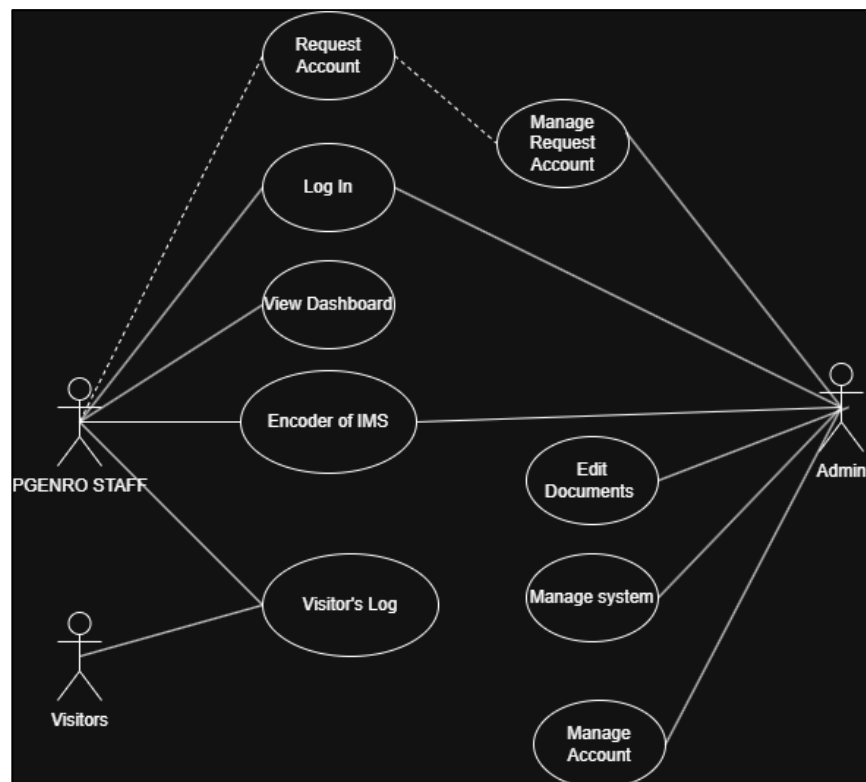


Figure 14. Use Case Diagram IMS

4.2. DFD Diagram

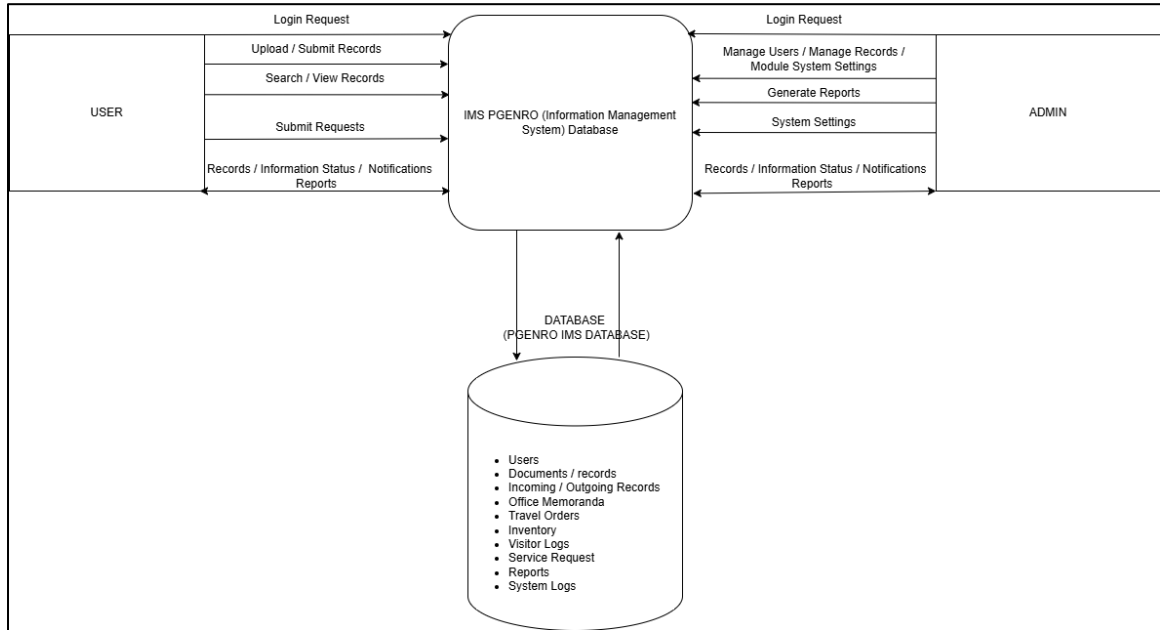


Figure 15. Data Flow Diagram IMS

Figure 15 illustrate the Data Flow Diagram (DFD) of the PGENRO Information Management System. The diagram will show how data will move between the users, administrator, system processes, and database. In this system, the user will be able to log in, submit or upload documents, access assigned records, view document status, and manage information based on their authorized role. The administrator will have wider access, allowing them to manage user accounts, verify document entries, update records, generate reports, and monitor system activities.

The DFD will also present the major processes of the system, such as user authentication, document management, automated document classification, digital repository storage, search and retrieval, and report generation. All submitted documents and records will be stored in the centralized database, which will serve as the main storage

for incoming and outgoing communications, travel orders, office memos, employee records, inventory, visitor's log, service requests, and Inventory Custodian Slips (ICS). Through this flow, the system will help reduce misplaced files, improve document tracking, organize records, and provide faster access to important information within PGENRO.

4.3. Activity Diagram

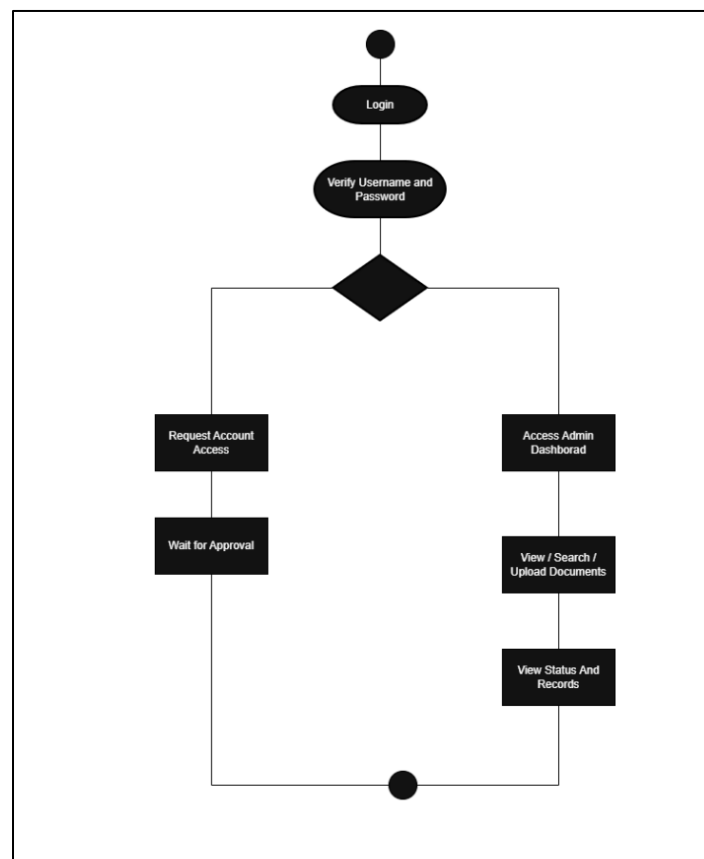


Figure 16. Activity diagram for user

Figure 16 portray the activity diagram designed for the user of the PGENRO Information Management System. This diagram will illustrate the sequence of activities to be performed by the user within the system, including logging in, accessing the dashboard, managing assigned modules,

uploading and viewing documents, searching records, and submitting requests for document updates or modifications. It will show how the user will interact with the system from authentication up to the completion of transactions, ensuring an organized and efficient flow of document and record management within PGENRO.

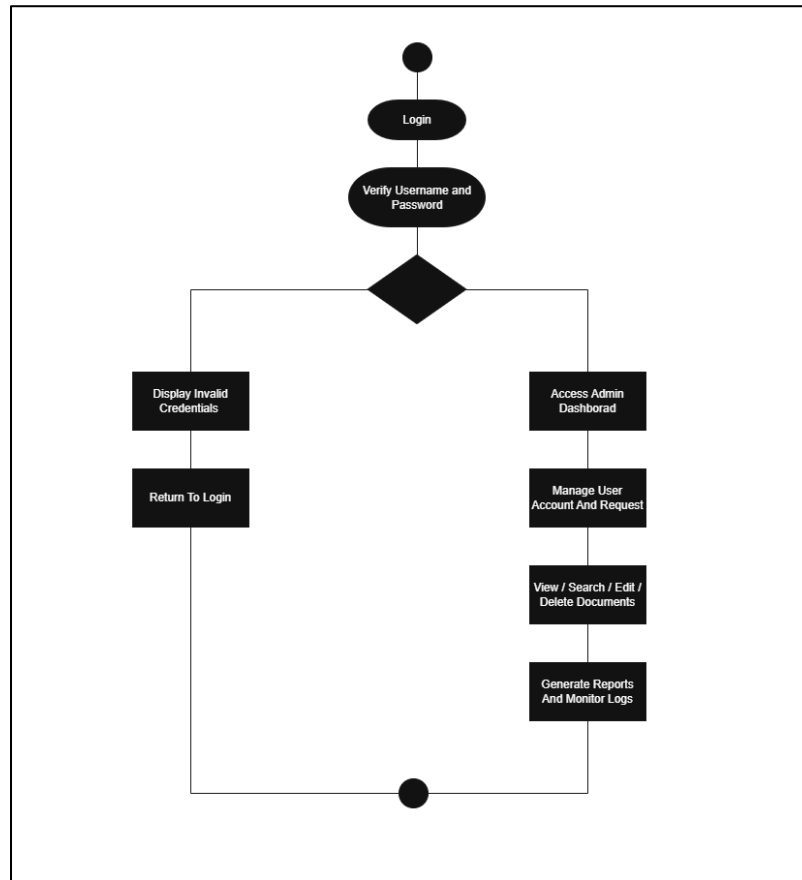


Figure 17. Activity diagram for admin

Figure 17 portray the activity diagram designed for the administrator of the PGENRO Information Management System. This diagram will illustrate the sequence of activities to be performed by the administrator within the system, including logging in, accessing the admin dashboard, managing user accounts, monitoring and processing records, approving or

updating documents, maintaining system modules, and generating reports.

It will present how the administrator will interact with the system from authentication to the completion of administrative tasks, ensuring effective control, organization, and management of records and system operations within PGENRO.

5. Development

The development of the enhanced PGENRO Information Management System will cover the complete process of system design, coding, testing, debugging, and deployment. This phase will focus on creating a web-based platform that can manage and organize PGENRO records such as incoming and outgoing communications, travel orders, office memos, employee records, inventory, visitor's log, service requests, and Inventory Custodian Slips.

The researchers will use appropriate web development tools and technologies such as Python, HTML, CSS, JavaScript, Visual Studio Code, and Firebase to build the system. These tools will serve as the foundation for developing the system interface, database, authentication, document uploading, and record management features. The development process will be guided by consultations with the technical adviser to ensure that the system design and functions are aligned with the objectives of the study.

During this phase, the researchers will encode the system modules, create the database structure, and develop the necessary functions for document storage, automated classification, user access control, searching, updating, editing, and deleting records. The system will also include security features such as user authentication and role-based access to ensure that only authorized users can access specific functions.

After the initial development, the researchers will conduct system debugging to identify and correct possible errors in the program. Any issues related to data entry, file uploading, document retrieval, classification, or user interface navigation will be reviewed and fixed. The system will then undergo retesting to ensure that the identified errors have been resolved and that the system performs according to its intended purpose.

The final part of the development phase will involve installing and deploying the system on the designated computer or web server. This will include setting up the database, configuring system files, creating user accounts, and uploading sample records and documents. This process will demonstrate how data and files are entered, stored, classified, retrieved, and managed within the enhanced PGENRO Information Management System.

6. Evaluation

The evaluation phase will determine the effectiveness, functionality, and acceptability of the enhanced PGENRO Information Management System. After the development and testing of the system, the researchers will conduct an evaluation to ensure that the proposed system meets the needs of PGENRO in managing records, documents, and office transactions.

The system will be evaluated using black box testing to check if each module functions according to its intended purpose. The testing will cover the major features of the system, including user login, document uploading, automated document classification, record searching, editing, deleting, generating control numbers, managing incoming and outgoing communications, travel orders, office memos, employee records, inventory, visitor's log, service requests, and Inventory Custodian Slips. The results will be recorded as either Passed or Failed based on the expected output of each function.

Aside from functionality testing, the system will also be evaluated by selected users using an evaluation questionnaire. The respondents may include PGENRO staff, administrators, and other authorized users who will assess the system based on usability, reliability, security, efficiency, and overall performance. Their responses will help determine whether the system is acceptable and useful for improving the office's information management process.

The gathered evaluation results will be analyzed using percentage computation and weighted mean. The percentage will be used to determine the success rate of the test cases, while the weighted mean will be used to interpret the acceptability rating of the system. A high evaluation result will indicate that the system is effective, functional, and suitable for implementation. If problems or low ratings are identified, the researchers will revise and improve the system before the final deployment.

7. Testing

The testing phase will be conducted to determine whether the enhanced PGENRO Information Management System functions correctly based on its intended features and objectives. This phase will help the researchers identify errors, system defects, and areas that need improvement before the final implementation of the system.

The researchers will use black box testing to evaluate the system's functionality without examining the internal source code. This testing will focus on the input and output of each module, including login authentication, request account access, document uploading, automated document classification, record searching, editing, deleting, control number generation, visitor's log management, inventory records, service requests, and Inventory Custodian Slip management.

During the testing process, the researchers will prepare test cases for each system function. Each test case will include the input data, expected output, actual output, and result. The result will be marked as Passed if the actual output matches the expected output, and Failed if the system does not perform the required function correctly.

If errors or failed test cases are found, the researchers will revise and debug the system. After correcting the identified issues, the affected modules will be tested again to ensure that the problems have been resolved. This process will continue until the system performs accurately and meets the required level of functionality.

Through this testing phase, the researchers will ensure that the enhanced PGENRO Information Management System is reliable, functional, and ready for evaluation by the intended users.

Data Analysis Plan

The researchers will use the Equivalence Partitioning technique to analyze the testing results of the enhanced PGENRO Information Management System. This technique will group the system inputs into valid and invalid categories, such as document types, metadata entries, and user roles. Through this method, the researchers can efficiently evaluate the system's functionality and user interface during black box testing without testing every possible input individually.

The results of the black box testing will be analyzed using percentage computation. The percentage will be used to determine the number of successful test cases compared to the total number of test cases executed. The formula to be used is:

$$\text{Percentage} = \frac{\text{Number of Successful Test Cases}}{\text{Total Number of Executed Test Cases}} \times 100$$

The computed percentage will serve as the basis for evaluating the overall performance of the proposed system. A success rate of 95% will be required for the system to be considered acceptable. If the system does not meet the required percentage, the researchers will revise, reprogram, and retest the system until the acceptable performance level is achieved.

In addition to the overall pass rate, the researchers will evaluate the automated classification component using standard classification metrics (precision, recall, F1) on labeled test sets to identify misclassification patterns and guide retraining. To determine the system's acceptability across ISO 25010 attributes such as compatibility, functional suitability, maintainability, flexibility, reliability, and interaction capability.

The researchers will use median and mode on Likert-scale questionnaire responses: the median will be found by ordering responses and selecting the middle value (or averaging the two middle values if the sample size is even), while the mode will be the most frequently occurring rating; these measures will summarize user perceptions and be compared against predefined acceptability cutoffs to decide whether further improvements are necessary.

Arithmetic Weighted Mean

The Arithmetic Weighted Mean (AWM) used to determine the overall acceptability of the proposed PGENRO Information Management System with Automated Document Classification and Digital Repository. It will be used to compute and interpret the responses of the evaluators based on the ISO/IEC 25010 software quality criteria.

The formula
be used is:

Symbol	Meaning
(WAM)	Weighted Arithmetic Mean
(f)	frequency of responses
(x)	assigned weight/rating value
$(\sum(fx))$	total weighted score
(N)	total number of responses/respondents

that will

Where:

Scale	Range	Verbal Interpretation
4	3.50 – 4.00	Highly Acceptable
3	2.50 – 3.49	Acceptable
2	1.50 – 2.49	Unacceptable
1	1.00 – 1.49	Highly Unacceptable

The respondents will evaluate the system using a four-point Likert scale. Each response will be assigned a corresponding numerical value to measure the acceptability of the system.

The computed AWM will be interpreted using the four-point Likert scale. This will help the researchers determine whether the proposed system will be rated as Highly Acceptable, Acceptable, Unacceptable, or Highly Unacceptable by the respondents.

ISO 25010 Criteria

The proposed PGENRO Information Management System with Automated Document Classification and Digital Repository will be evaluated using the ISO/IEC 25010 Software Quality Model. The evaluation will include nine parameters: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, portability, and interaction capability. These criteria will serve as the basis for determining the quality, effectiveness, and acceptability of the developed system.

Functional Suitability

Functional suitability refers to the ability of the system to provide the necessary functions required by PGENRO. This parameter will evaluate whether the system can properly manage incoming and outgoing communications, travel orders, office memos, employee records, inventory, visitor's log, service requests,

and Inventory Custodian Slips. It will also assess the effectiveness of the automated document classification, digital repository, search and retrieval, and report generation features.

Performance Suitability

Performance efficiency refers to the ability of the system to perform its functions within an acceptable time and resource usage. This parameter will evaluate the speed and responsiveness of the system during login, document uploading, classification, data entry, searching, retrieving records, and generating reports. It will also assess whether the system can process transactions without unnecessary delays.

Usability

Usability refers to the ease of learning, understanding, and operating the system. This parameter will evaluate the clarity of the interface, dashboard, menus, buttons, forms, labels, and instructions. It will also assess whether administrators and authorized users can perform their tasks with minimal difficulty.

Reliability

Reliability refers to the ability of the system to perform consistently and correctly under normal operating conditions. This parameter will evaluate whether the system can maintain accurate records, prevent data loss, handle errors, and remain available when needed. It will also assess the system's consistency in storing, retrieving, and updating PGENRO records.

Security

Security refers to the ability of the system to protect confidential records

and documents from unauthorized access. This parameter will evaluate the system's secure login, role-based access control, user account verification, audit trail, and protection of stored documents in the digital repository. It will also assess whether only authorized users can access, edit, delete, or manage specific records.

Maintainability

Maintainability refers to the ability of the system to be corrected, updated, modified, or improved when needed. This parameter will evaluate whether the system can be maintained by future developers or administrators. It will also assess how easily errors can be fixed, modules can be updated, and additional features can be integrated without affecting the entire system.

Portability

Portability refers to the ability of the system to be transferred or used in another compatible environment with minimal changes. This parameter will evaluate whether the web-based system can operate on different computer units, browsers, or hosting environments. It will also assess whether the system can be deployed or transferred without major modifications.

Interaction Capability

Interaction capability refers to the ability of the system to support clear, effective, and efficient interaction between the users and the system. This parameter will evaluate whether the system provides understandable navigation, proper prompts, feedback messages, confirmation alerts, and error notifications. It will also assess whether the system allows administrators and authorized users to interact with the modules smoothly, especially in document uploading,

classification, searching, editing, and report generation.

Implementation Plan

Table 2

Implementation Plan and Strategies of the System

Strategy	Activities	Persons Involved	Duration
Approval from PGENRO Office	Secure endorsement and approval from PGENRO Officials	Proponents PGENRO Head	2 Days
System Deployment	Installation of the web-based system on authorized servers, Firebase/hosting setup	Proponents; MIS Personnel	1 Day
Documentation Turnover	Provide manuals, license, source code, and maintenance plan	Proponents; MIS Personnel	1 Day
User Orientation	Conduct brief orientation, distribute user guides	Proponents; Division Heads; Staff	1 Day
User Training	Hands-on demonstration on system usage and data entry procedures	Proponents; Division Heads; Staff	1 Day

Ethical Consideration

This study ensures adherence to ethical principles to protect respondents' rights, maintain integrity, and uphold research standards in the enhancement of the PGENRO Information Management System. All participants will be provided with a clear explanation of the study's objectives, procedures, and their role; informed consent will be obtained prior to their involvement, and participation will be voluntary with the option to withdraw at any time without penalty. Respondent identities, sample documents, and feedback will be safeguarded to prevent unauthorized access or disclosure; personal and sensitive data used for testing will be minimized, anonymized or pseudonymized when possible, and handled with strict confidentiality. Access to test data and the live system will be controlled through role-based permissions, and data will be protected by encryption in transit and at rest, together with secure audit logging to ensure accountability.

Because the proposed enhancement may include biometric features for visitor logging, biometric data will be treated as highly sensitive: explicit consent will be required for capture, encrypted storage and defined retention policies will be enforced, and an alternative non-biometric logging method will be provided for those who decline. Evaluators and end-users will be selected fairly and based on relevance to the study to ensure balanced representation; recruitment will avoid conflicts of interest and participants will receive adequate instructions and training to perform evaluations without risk of professional or emotional harm. Test scenarios will avoid disrupting PGENRO operations; where production data is necessary, it will be sanitized or replaced with synthetic datasets to prevent operational impact.

The researchers will report findings, methodologies, and system evaluations truthfully, avoiding bias, fabrication, or manipulation of results, and will properly cite all

software frameworks and scholarly works used to uphold academic integrity. The project will comply with Southern Luzon State University research policies and applicable national standards and maintenance plan will be provided at project handover specifying responsibilities for taxonomy stewardship, access management, retention schedules, incident response, and procedures for addressing ethical concerns that may arise after deployment.

Reference Citation

- Laudon, K. C., & Laudon, J. P. (2016). *Management Information Systems: Managing the Digital Firm*. Pearson.
<https://books.google.com.ph/books?id=4RAwDwAAQBAJ&hl=fil>
- Del Sole, A. (2023). *Visual Studio Code distilled: Evolved code editing for Windows, macOS, and Linux* (2nd ed.). press. <https://doi.org/10.1007/978-1-4842-9484-0>
- Alshamaa, Z. M., & Saleem, N. N. (2025). Black box software testing techniques: A literature review. *Passer Journal of Basic and Applied Sciences*, 7(2), 1001–1011.
<https://doi.org/10.24271/psr.2025.508216.1963>
- Santi, P. A. D. A., Afwani, R., Albar, M. A., Anjarwani, S. E., & Mardiansyah, A. Z. (2022). Black box testing with equivalence partitioning and boundary value analysis methods (Study case: Academic Information System of Mataram University). In *Proceedings of the 1st International Conference on Management, Business, Innovation, Education and Social Science (MIMSE-I-C 2022)* (pp. 207–219). Atlantis Press. https://doi.org/10.2991/978-94-6463-084-8_19
- Nugraha, U., & Rachman, M. A. N. (2024). *Evaluation of Ixitask Application Quality Based on ISO/IEC 25010:2023 in Marketing Division*. Atlantis Press.
<https://www.atlantis-press.com/proceedings/wicoeng-24/126007237>
- Hadiapurwa, A., et al. (2024). *Assessment of the Museum Pendidikan Nasional website using the System Usability Scale*. Atlantis Press.
<https://www.atlantis-press.com/article/126005273.pdf>

- Madaminov, U. A., & Allaberganova, M. R. (2023). *Firestore Database Usage and Application Technology in Modern Mobile Applications*. IEEE conference listing / ResearchGate entry with DOI shown.
https://www.researchgate.net/publication/376494147_Firebase_Database_Usage_and_Application_Technology_in_Modern_Mobile_Applications
- Rashmi, Baba, A. A., & Bhagwan, L. (2021). *A Study the basic Programming Language of Hypertext Markup Language*. Journal of Advances in Science and Technology
https://www.researchgate.net/publication/381696973_A_Study_the_basic_Programming_Language_of_Hypertext_Markup_Language
- Jain, R., Shrivastava, V., Pandey, A., & Sharma, A. (2024). *Modern web development using CSS & HTML*. *International Journal of Emerging Science and Engineering*, 12(6), 13–16.
<https://journals.blueeyesintelligence.org/index.php/ijese/article/view/379>
- Bhanarkar, N., Paul, A., & Mehta, A. (2023). *Responsive web design and its impact on user experience*. *International Journal of Advanced Research in Science, Communication and Technology*, 3(4).
https://www.researchgate.net/publication/370134359_Responsive_Web_Design_and_Its_Impact_on_User_Experience
- Malaya, A. R. N., Munar, E. A., Cuison, F. P., & Dacanay, E. G. (2022). *Information management system for research of Don Mariano Marcos Memorial State University–South La Union Campus*. *Indonesian Journal of Electrical Engineering and Computer Science*, 28(3), 1668–1675.
<https://doi.org/10.11591/ijeecs.v28.i3.pp1668-1675>
- Llácer Luna, E., Garigliotti, D., Martínez-Plumed, F., & Ferri Ramírez, C. (2024). Automatic PDF document classification with machine learning. In *Intelligent Data Engineering and Automated Learning – IDEAL 2024: 25th International Conference, Proceedings, Part I* (pp. 447–459). Springer.
https://doi.org/10.1007/978-3-031-77731-8_40
- Pandey, M., Arora, M., Arora, S., Goyal, C., Gera, V. K., & Yadav, H. (2023). AI-based integrated approach for the development of intelligent document management system (IDMS). *Procedia Computer Science*, 230, 725–736.
<https://www.sciencedirect.com/science/article/pii/S1877050923021324?via%3Dihub>

